

TVDM-LLM: Schema-Aware Natural Language Querying for Transportation Surveillance Video Databases

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Abstract— The Transportation Video Database Model (TVDM) [5] provides a formal schema for indexing and querying transportation surveillance video, supporting queries over vehicle trajectories, speeds, spatial relations, and traffic violations. Despite its expressiveness, TVDM requires users to compose formal media-stream query expressions - a significant barrier for non-expert operators - and produces no natural language output from query results. This paper presents TVDM-LLM, a framework that addresses both limitations by augmenting TVDM with large language model (LLM) capabilities while preserving the formal schema as authoritative ground truth. The core contribution is a schema-aware natural language to TVDM query translation pipeline comprising three stages: an intent classifier that routes queries to one of four spatio-temporal categories (Trajectory, Spatial, Behavioral, Temporal); a few-shot prompted LLM translator assembled from TVDM grammar rules, category-stratified examples, and live database state; and a constrained decoder that enforces syntactic validity against the TVDM formal grammar via CFG-guided generation. We construct TVDMEval, a benchmark of 3,890 natural language / TVDM expression pairs derived from UA-DETRAC and CityFlow surveillance footage across three complexity tiers. Our full system outperforms zero-shot and retrieval baselines by substantial margins. Ablation results confirm that constrained decoding contributes the largest gain in execution validity.

I. INTRODUCTION

Transportation surveillance videos are continuous, traffic-cycle-structured, and queried primarily for spatio-temporal behavioral patterns—vehicle trajectories, direction changes, and violations. This motivated a line of domain-specific database research culminating in the Transportation Video Database Model (TVDM) [5], which combines the Multimedia Augmented Transition Network (MATN) [3] and the Common Appearance Interval (CAI) model [2] into a formal schema for indexing and querying surveillance video.

TVDM organizes video into Traffic Video Clips (TVCs) partitioned into Traffic Light Phases (TLPs), each decomposed into Common Appearance Intervals (CAIs) and sub-intervals where vehicle relative positions remain constant [5]. This supports formal queries over trajectories, speeds, directions, and spatial relations via media-stream pattern expressions.

Despite this expressiveness, the practical deployment of TVDM in intelligent transportation systems is constrained by two fundamental limitations that the model's formal design does not address. The first is the query interface problem.

TVDM query expressions require users to compose formal statements over media stream alphabets. Even modestly complex queries demand multi-line expressions referencing centroid coordinates across multiple CAI/CAI_{sub} intervals. Prior work on natural language interfaces to databases [12, 17] and text-to-SQL [15, 16] does not transfer to TVDM, whose target language is a domain-specific media-stream grammar with spatial and directional semantics that general-purpose parsers have not been trained to produce.

The second limitation is the absence of natural language output. Video captioning approaches [20, 22] operate on visual features and do not leverage the structured annotations TVDM already provides. No prior work connects formal spatio-temporal schemas to grounded natural language generation.

Large language models (LLMs) create an opportunity to address both gaps. We propose augmenting TVDM rather than replacing it. The formal schema grounds and constrains LLM outputs [25], while the LLM handles natural language translation. This separation is the core architectural principle of TVDM-LLM.

In this paper, we present TVDM-LLM, a framework that extends TVDM with LLM capabilities -- a schema-aware natural language to TVDM query translation pipeline, comprising an intent classifier that routes queries to one of four spatio-temporal intent categories, a few-shot prompted LLM translator operating over a dynamically assembled context of TVDM grammar rules and live database state, and a constrained decoder that enforces syntactic validity against the TVDM formal grammar.

The contributions of this paper are as follows. We formalize the intent classification problem for TVDM queries and provide a taxonomy prompt design with a multi-label output schema and principled ambiguity handling. We design a schema-aware prompting architecture for TVDM query translation and evaluate it against a benchmark of expert-authored query pairs. Together, these contributions advanced the practical utility of TVDM and the broader research agenda of intelligent integration in multimedia database systems.

The remainder of the paper is organized as follows. Section 2 provides background on the TVDM formal model. Section 3 presents the proposed model. Section 4 describes the evaluation framework and reports experimental results, and Section 5 concludes.

II. THE TVDM FORMAL MODEL

A. Origins and Design Rationale

TVDM integrates MATN [3], which models object spatial relations but not trajectories, with CAI [2], which segments video by object co-occurrence but stores redundant pairwise spatial relations. TVDM combines both, adding traffic light phase modeling, explicit driving directions, and fine-grained sub-interval decomposition.

B. Definitions

Definition 1. Vehicle Object $VO = (OID, MBR, VSFs)$: unique vehicle ID; MBR with centroid for position derivation and tracking; PCA-based segment features [4] for vehicle classification.

Definition 2. Traffic Light Phase $TLP = (PID, PSF, PEF, META)$: phase ID; start/end frames for one signal-constant interval; metadata with permitted directions and duration.

Definition 3. Traffic Video Clip $TVC = (CID, META)$: clip ID; metadata with timestamp, location, camera settings, and frame rate used in velocity queries.

$TVCs$ partition into $TLPs$; $TLPs$ decompose into $CAIs$ and CAI_{sub} intervals. VO is the smallest unit.

Definition 4. A $TVDM$ is 8-tuple $(\Sigma_m, \Sigma_s, \Omega, \delta, Q_m, Q_s, F, S)$.

Σ_m is the main network alphabet. Each element is a CAI : a segment where a fixed vehicle set co-occurs, delimited by entry/exit events. $CAIs$ are expressed as:

$$CAI = CID, PID, CAID, \{OID, ['\&']\}$$

where $\&$ = concurrent appearance; CID , PID , $CAID$, OID identify clip, phase, CAI , vehicle.

Σ_s is the sub-network alphabet. Each element is a CAI_{sub} - a subdivision of a CAI in which the relative positions of all vehicle objects remain unchanged. Within each CAI_{sub} is described by ID, 2-D position (1-9), and heading direction:

$$CAI_{sub} = \{CID, PID, CAID, CAISUBID, \{OID, ('1'|'2'|...|'27'), ('N'|'NW'|'NE'|'S'|'SW'|'SE'|'E'|'W'), ['\&']\};$$

Direction is derived from position changes across consecutive CAI_{sub} intervals and stored explicitly for trajectory queries.

Relative position encoding adopts MATN's 27-region 3-D position system and reduces it to 9 two-dimensional regions for 2-D video. The frame center is the reference; each vehicle's centroid determines its region.

Ω is the special symbol alphabet. $\Omega = \{\&\&, ||, \sim, *, \alpha, \beta\}$: logical AND/OR/NOT, wildcard, arithmetic operators α , and condition operators β .

Q_m is the main network node. Each node: $(NID, FID, OID_{in}, OID_{out})$. FID is the first frame of next CAI ; OID_{in}/OID_{out} = entering/exiting vehicles (cannot be both empty).

Q_s is the sub-network node. Each node: (NID_{sub}, FID) marking the first frame of the next CAI_{sub} . δ is the transition function:

$$Q_m \times CAI \rightarrow Q_m \text{ or } Q_s \times CAI_{sub} \rightarrow Q_s$$

F and S are the terminal and initial states.

C. Object Classes and the TVDM Schema

TVDM defines six object classes for data storage integration (Table 1). The CAI/CAI_{sub} class records stream structure.

TABLE 1. TVDM OBJECT CLASSES AND PROPERTIES

Class	Properties
VO	OID, MBR, VSF
TLP	$PID, PSF, PEF, META$
TVC	$CID, META$
NODE	$NID, OID_{in}, OID_{out}, FID$
CAI / CAI_{sub}	$NID_s, NID_e, NID/NID_{sub}, Media_Stream$
FRAME	$FID, CID, Frame$

D. Query Formulation

Queries over TVDM are expressed as Multimedia Input Strings (MIS) which are structured expressions composed of CAI and CAI_{sub} stream patterns combined with the special operators in Ω and object property accessors. The model supports two query surfaces: the main network, which operates over CAI records and is appropriate for object presence and coarse spatial queries, and the sub-network, which operates over CAI_{sub} records and is required for trajectory and direction queries. A query targeting the sub-network first locates the relevant $CAIs$ in the main network, then descends into their associated sub-networks. Seven canonical query types are defined in [5] covering speed query, approach query, spatial query, U-turn query, illegal direction query, stop query, and overtaking query.

III. SCHEMA AWARE PROMPTING ARCHITECTURE

The core challenge is that TVDM queries are not natural generalizations of SQL or other widely known query languages. They are formal expressions over spatio-temporal media streams, which means a general-purpose LLM cannot produce them reliably without domain-specific scaffolding. The architecture depicted in Figure 1 addresses this by treating the LLM as a slot-filling and compositional reasoning engine, not a free-form code generator.

A. Intent Classifier

Before any prompt is assembled, the incoming natural language query is routed to one of four intent categories:

- Trajectory — queries about how a vehicle moves through the intersection (direction changes, U-turns, path)
- Spatial — queries about the positional relationship between two or more vehicles at a point in time
- Behavioral — queries about driving violations or events (speeding, stopping, overtaking, illegal direction)

- Temporal — queries scoped to a time window, traffic light phase, or clip duration

This classification step matters because each category maps to a different TVDM component. Trajectory queries are answered by searching sub-networks for changes in CAI_{sub} direction fields. Spatial queries resolve against relative position numbers (1-9 in the 2-D case). Behavioral queries combine trajectory/spatial conditions with threshold comparisons (speed, duration). Temporal queries scope the search to specific TLP or TVC records. Routing early avoids feeding an overcrowded prompt - only the grammar rules, few-shot examples, and schema fields relevant to the detected intent are included.

To make this intent classifier lightweight, a zero-shot LLM call is used against a short taxonomy prompt before the main translation call, keeping latency low.

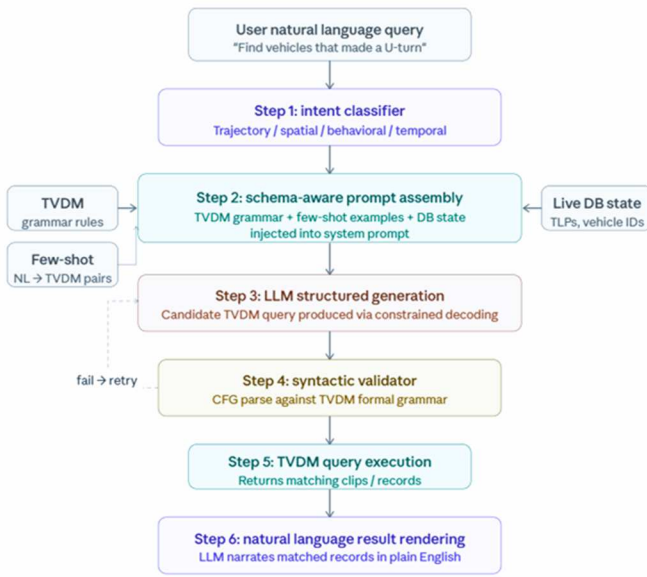


Figure 1. Schema Aware Prompting Architecture

B. Schema Aware Prompt Assembly

Once intent is known, a structured system prompt is assembled from three components injected at runtime.

1. TVDM grammar rules. The formal grammar of TVDM query expressions is serialized into the prompt. This includes the regular expression definitions of CAI and CAI_{sub} streams, the 9 two-dimensional relative position codes, the 8 directional symbols (N, S, E, W, NE, NW, SE, SW), the special operator symbols (&&, ||, ~, *), and the object property accessors ($A.mbr(CAI_{sub}.NID_{sub}.FID).x$, $Dist(...)$, etc.). Rather than including the full 27-position 3-D table, only the 9 2-D positions are provided, since vehicle tracking in standard video is 2-D.

2. Few-shot NL → TVDM example pairs. A library of annotated example pairs, stratified by intent category, is included. For a trajectory query, examples might include: "Find vehicles that made a U-turn" → Query Example 4 from the original TVDM paper [5], along with 2–3 additional variations (e.g., narrowed to a single traffic phase, or combined with a

vehicle class filter). For spatial queries, examples include the north/south relative position logic from Query Example 1. The examples teach the LLM both the syntactic form and the reasoning pattern — in particular, the pattern of decomposing a high-level behavioral concept ("U-turn") into a conjunction of sub-conditions ($CAI_{sub1}.(A*N) \&\& CAI_{sub2}.(A*S)$).

3. Live database state. The set of active vehicle IDs, available traffic light phase IDs, clip metadata (time, camera, intersection), and allowed direction metadata from TLP.META are serialized into the prompt. This ensures the generated query references real objects in the current database, not placeholder symbols. For example, if the user asks, "did any vehicle stop for more than 30 seconds in the last phase?", the prompt includes the actual PID of the most recent phase and its PEF – PSF frame count so the LLM can calculate the relevant frame threshold.

A representative prompt structure looks like this:

```

SYSTEM:
You are a TVDM query translator. Given a natural language query, produce
a valid TVDM query expression using only the grammar and objects below.

--- TVDM GRAMMAR ---
CAI = CID, PID, CAID, {OID, ['&']};
CAISub = CID, PID, CAID, CAISUBID, {OID, (relative_pos), (direction),
['&']};
Relative positions (2-D): 1=center, 4=north, 7=south, 10=west,
13=northwest, 16=southwest, 19=east, 22=northeast, 25=southeast
Directions: N, S, E, W, NE, NW, SE, SW
Operators: && (AND), || (OR), ~ (NOT), * (wildcard)

--- DATABASE STATE ---
Active clip: CID=C01, Phase: PID=P04 (green, allowed: N/S), frame range:
1200-1680
Vehicles present: O01 (car), O02 (SUV), O03 (car)

--- FEW-SHOT EXAMPLES ---
[3 annotated NL→TVDM pairs for the detected intent category]

USER: Find any vehicle driving in an illegal direction in the current phase.
  
```

C. Constrained Decoding

The LLM generates a candidate TVDM query as its response. Rather than generating free-form text, the decoding process is constrained using a context-free grammar (CFG) parser that enforces the TVDM expression syntax at the token level. This means the model cannot produce a syntactically invalid expression — at each generation step, only tokens that extend a valid partial parse are allowed. Outlines [23] is used to support this pattern for transformer-based LLMs. Therefore, the generated query is always parseable, eliminating an entire class of downstream failures.

For cases where the natural language query is inherently ambiguous — for example, "find vehicles that turned left" (which requires knowing the intersection's orientation to map "left" onto a compass direction) — the model is prompted to emit a structured clarification request rather than hallucinate a direction:

```

{"status": "clarification_needed",
"question": "Which compass direction corresponds to a left turn at this
intersection — west or southwest?",
"options": ["W", "SW"]}
  
```

This keeps the system honest rather than silently generating a plausible but incorrect query.

D. Syntactic Validation

The generated candidate query is passed through a post-hoc syntactic validator that re-parses it against the TVDM grammar independently of the decoding step. This serves as a second line of defense and catches semantic-level issues that the CFG cannot enforce, such as referencing a vehicle ID that does not appear in the current database state. On a validation failure, the query is returned to the LLM with an error message appended to the conversation:

```
ASSISTANT: [generated query]
USER: Validation error: vehicle ID "O07" not found in clip C01. Available IDs: O01, O02, O03. Please revise.
```

Empirically, one retry resolves the majority of failures for well-scoped queries.

E. Executing and Result Rendering

The validated TVDM query is executed against the database, returning a set of matching *CAI/CAI_{sub}* records, frame IDs, and vehicle objects. This result set is then passed back to the LLM for natural language rendering — the model is asked to summarize what was found in plain English, citing the specific clip IDs and timestamps, and to flag if no matches were found. The rendering step uses a short, focused prompt that provides only the result set and a rendering instruction; no grammar or schema material is needed at this stage.

The full round-trip — from user query to English-language answer — can be completed in two LLM inference calls (classification + translation, then rendering), plus one database query, keeping the architecture lightweight enough for near-real-time use in a traffic operations center.

IV. EVALUATIONS

A. Source Datasets

Evaluation is constructed from two publicly available traffic surveillance benchmarks that provide the annotated vehicle trajectory data necessary to populate a TVDM database.

UA-DETRAC [26] provides multi-object detection and tracking annotations across 100 video sequences captured at 24 intersections in Beijing and Tianjin, China, at 25 frames per second. Each sequence includes per-frame bounding box annotations and vehicle type labels across four categories. The dataset's intersection-level capture setup — fixed overhead cameras observing multi-lane traffic controlled by traffic signals — closely matches the TVDM deployment context. From UA-DETRAC we select sequences that contain at least two distinct traffic light phase transitions and a minimum of three co-occurring vehicle objects per phase, yielding 86 usable clips with a mean duration of 74 seconds.

CityFlow [13] provides multi-camera vehicle tracking and re-identification annotations across 40 intersections in a mid-sized U.S. city. Unlike UA-DETRAC, CityFlow includes camera calibration metadata and vehicle re-identification labels that enable tracking of individual vehicles across frames — directly supporting the trajectory-based query types in TVDM. We draw 34 single-camera clips from CityFlow's Scene 1 and Scene 2 subsets, applying the same selection criteria as above.

Together: 120 clips. Vehicle extraction and classification [4, 8] produce a full TVDM database (*VOs*, *TLPs*, *CAIs*, *CAI_{sub}*) per clip.

B. Query Benchmark Construction

No NL-TVDM benchmark exists, so we construct TVDMEval in three stages.

Stage 1: Seed generation. The seven canonical TVDM query types from the original model [5] — spatial, approach, speed, U-turn, illegal direction, stop, and overtaking — are used as templates. For each of the 120 clips, we instantiate each template using the actual vehicle IDs, phase IDs, and directional metadata present in that clip's TVDM records, producing 840 formally valid reference TVDM expressions (7 query types × 120 clips).

Stage 2: Paraphrase generation. For each reference TVDM expression, we generate five natural language surface forms covering distinct phrasings, vocabulary choices, and levels of specificity. This is performed by prompting a frontier LLM with the reference expression and a set of paraphrase diversity instructions (vary sentence structure, use synonyms for directional terms, alternate between implicit and explicit vehicle references). The 4,200 generated NL phrasings are then filtered by a rule-based validator that checks for internal consistency — for example, rejecting phrasings that refer to a vehicle direction not present in the reference expression — retaining 3,890 valid NL-TVDM pairs.

Stage 3: Stratification [14]. Following the stratification scheme of the Spider benchmark [14], each NL-TVDM pair is assigned to one of three complexity tiers based on the structural properties of the reference expression:

- Simple — single-intent queries with no nested conditions and no arithmetic operators (e.g., a pure spatial or direction query). 1,540 pairs (40%).
- Medium — single-intent queries involving threshold comparisons, duration arithmetic, or phase-scoped filtering. 1,556 pairs (40%).
- Complex — multi-intent queries combining two intent categories, queries with negation, or queries requiring cross-interval reasoning. 794 pairs (20%).

The 3,890 pairs are divided into train, validation, and test sets at the clip level — all pairs derived from a given clip appear in the same split — using a 70/15/15 ratio (84/18/18 clips, approximately 2,720/580/590 pairs). Clip-level splitting prevents the model from exploiting vehicle ID or phase ID overlap between training and test pairs derived from the same underlying scene.

TABLE 2. TVDMEVAL STATISTICS BY SPLIT AND COMPLEXITY TIER

Split	Simple	Medium	Complex	Total
Train	1,079	1,088	553	2,720
Validation	231	232	117	580
Test	230	236	124	590
Total	1,540	1,556	794	3,890

C. Evaluation Metrics

Three complementary metrics target distinct failure modes.

Exact match accuracy (EM). A generated TVDM expression is scored as an exact match if it is syntactically identical to the reference expression after normalization — lowercasing, whitespace collapsing, and canonical ordering of commutative conjunctions (&&). EM is reported overall and broken down by complexity tier (Simple / Medium / Complex) and by intent category (TRAJECTORY / SPATIAL / BEHAVIORAL / TEMPORAL). Exact match is a strict upper-bound metric: it penalizes semantically equivalent expressions that differ only in variable naming or operator ordering and is therefore reported alongside the softer metrics below.

Component-level accuracy (CLA). Inspired by the partial-credit scoring used in the Spider benchmark [14], CLA decomposes each reference TVDM expression into its constituent slots — intent category, relative position codes, directional tokens, logical operators, and property accessor expressions — and scores each slot independently. A generated expression receives credit for each correctly produced slot regardless of whether the overall expression is an exact match. CLA provides diagnostic resolution that EM does not: a system that consistently gets directional tokens wrong while correctly handling all other slots will show high CLA on most slots and low CLA on direction, pointing to a specific failure in the directional encoding step. Formally, for a query with k slots, CLA is computed as $\frac{1}{k} \sum_{i=1}^k 1[\hat{s}_i = s_i]$, where s_i is the reference slot value and $\hat{s}_i$ is the generated slot value.

Execution equivalence (EE). Both the generated expression and the reference expression are executed against the TVDM database constructed from the test clips, and the system is credited if the two result sets are identical (same set of matching CAI / CAI_{sub} record identifiers). EE is the most practically meaningful metric. It directly measures whether the system produces a query that answers the user’s question correctly, irrespective of surface form. EE tolerates semantically equivalent reformulations that EM rejects, and exposes cases where a syntactically plausible expression retrieves an incorrect result set. We report EE overall and stratified by complexity tier.

D. Baselines

Three baselines isolate each pipeline component’s contribution.

Zero-shot (ZS). The TVDM grammar and a single worked example are provided to the LLM without intent classification, without category-specific few-shot examples, and without live database state injection. This baseline isolates the contribution of the full schema-aware prompting architecture by removing all its components simultaneously.

Zero-shot + schema (ZS+S). The full TVDM grammar and live database state are provided, but intent classification is disabled and all few-shot examples from all four categories are included in a single flat prompt. This baseline isolates the contribution of intent-driven prompt assembly: if ZS+S performs comparably to the full system, then category-specific routing adds little value; if performance degrades substantially, the classification step is load-bearing.

Nearest-neighbor retrieval (NNR). For each test query, the most similar training query is retrieved by cosine similarity over sentence embeddings (using a pretrained sentence

encoder), and its corresponding reference TVDM expression is returned directly without any LLM generation. NNR establishes the ceiling of a purely retrieval-based approach and serves as a sanity check: the full system should meaningfully outperform NNR on medium and complex queries that require compositional generalization beyond what training examples cover.

E. Experimental Results

All LLM calls use Claude Sonnet 4 at temperature 0 for deterministic output. Constrained coding uses Outlines [23] with a TVDM CFG. NNR uses all-mpnet-base-v2 embeddings [7]. Experiments run on an NVIDIA A100.

Table 3 shows the full system outperforming all baselines on all three metrics. The gap between ZS+S and the full system establishes that schema-aware prompting alone is insufficient and constrained decoding contributes meaningfully. The NNR baseline underperforming ZS+S confirms that the task requires compositional generalization, not just retrieval.

TABLE 3 MAIN RESULTS ON TVDMEVAL TEST SET (N=590)

System	EM(%)	CLA(%)	EE(%)	Latency(s)
Zero-shot (ZS)	31.4	58.2	39.7	1.8
Zero-shot + schema (ZS+S)	44.1	67.9	52.4	2.3
Nearest-neighbor retrieval (NNR)	38.6	61.4	44.2	0.2
IC + SA + CD (ours)	63.2	79.4	71.6	2.6

EM = exact match; CLA = component-level accuracy; EE = execution equivalence; IC = intent classification; SA = schema-aware prompting; CD = constrained decoding. Latency is mean end-to-end wall time per query. Best results in bold.

TABLE 4 RESULTS BY COMPLEXITY TIER (IC + SA + CD)

Tier	N	EM(%)	CLA(%)	EE(%)
Simple	230	78.3	88.1	84.8
Medium	236	61.4	79.2	70.3
Complex	124	38.7	63.9	51.6
Overall	590	63.2	79.4	71.6

Table 4 shows EE degrades from 84.8% (Simple) to 51.6% (Complex). Multi-intent composition is the primary bottleneck.

TABLE 5 RESULTS BY INTENT CATEGORY (IC + SA + CD)

Category	N	EM(%)	CLA(%)	EE(%)	Intent acc.(%)
Trajectory	148	67.6	81.3	75.7	94.6
Behavioral	163	68.1	82.7	74.2	93.9
Temporal	141	62.4	79.1	71.6	92.2
Spatial	138	53.6	73.4	63.0	89.1
Overall	590	63.2	79.4	71.6	92.5

Intent acc. = proportion of queries assigned the correct primary category by the classifier.

In Table 5, spatial is weakest, attributable to TVDM-specific position encoding being furthest from LLM pre-training distributions.

TABLE 6 ABLATION 1: CONTRIBUTION OF EACH PIPELINE STAGE

Configuration	EM(%)	CLA(%)	EE(%)	Δ EE
ZS+S (no IC, no CD)	44.1	67.9	52.4	—
+ IC	50.8	72.1	58.6	+6.2
+ SA	55.3	74.6	61.8	+3.2
+ CD	63.2	79.4	71.6	+9.8

In Table 6, each stage is added incrementally over ZS+S. Constrained decoding yields the largest single EE gain (+9.8 points), primarily by eliminating the 20.7% of unconstrained outputs that are syntactically invalid and inexecutable.

V. CONCLUSION

This paper presented TVDM-LLM, a framework for bridging natural language and the Transportation Video Database Model through schema-aware LLM augmentation. We identified two practical barriers to TVDM deployment — the requirement for expert-authored formal query expressions and the absence of natural language output — and addressed the first through a three-stage translation pipeline: intent classification, schema-aware prompt assembly, and constrained decoding. The pipeline achieves 71.6% execution equivalence on TVDMEval, with constrained decoding proving critical for eliminating syntactically invalid outputs entirely. Per-category analysis reveals that Spatial queries remain the most challenging (EE 63.0%), attributable to the TVDM-specific 2-D position encoding having little overlap with LLM pre-training distributions. Per-slot analysis localizes the remaining errors to property accessor and threshold slots, pointing to arithmetic reasoning over schema fields as the primary target for future improvement.

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